WORKED SOLUTION

STEP II 1998, Question 3

Pure • Telescoping Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q3

The key idea

A general term with a product of consecutive factors underneath begs to be split into partial fractions. The pieces then cancel across the sum (they telescope), leaving only a few terms at the two ends. In the second part, a ratio a_n/a_{n-1} means a_n/a_1 is a product of ratios, and that product telescopes too. With luck it turns a_n back into the series from the first part, which it does.

Solution

The sum S_N

Write

$$\frac{2n-1}{n(n+1)(n+2)} = \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2}.$$

Using the cover-up rule: at $n=0$, $A = \frac{-1}{1\cdot 2} = -\frac{1}{2}$; at $n=-1$, $B = \frac{-3}{(-1)(1)} = 3$; at $n=-2$, $C = \frac{-5}{(-2)(-1)} = -\frac{5}{2}$. So

$$\frac{2n-1}{n(n+1)(n+2)} = -\frac{1}{2n} + \frac{3}{n+1} - \frac{5}{2(n+2)}.$$

Notice that $A+B+C=0$. That is what makes the sum telescope.

Let $H_N = 1 + \frac{1}{2} + \cdots + \frac{1}{N}$. Then

$$\sum_{n=1}^{N} \frac{1}{n+1} = H_N - 1 + \frac{1}{N+1}, \qquad \sum_{n=1}^{N} \frac{1}{n+2} = H_N - 1 - \frac{1}{2} + \frac{1}{N+1} + \frac{1}{N+2}.$$

Hence

$$\begin{aligned} S_N &= -\tfrac{1}{2}H_N + 3\left(H_N - 1 + \frac{1}{N+1}\right) - \tfrac{5}{2}\left(H_N - \tfrac{3}{2} + \frac{1}{N+1} + \frac{1}{N+2}\right) \\ &= \left(-\tfrac{1}{2} + 3 - \tfrac{5}{2}\right)H_N + \left(-3 + \tfrac{15}{4}\right) + \left(3 - \tfrac{5}{2}\right)\frac{1}{N+1} - \frac{5}{2(N+2)} \\ &= \frac{3}{4} + \frac{1}{2(N+1)} - \frac{5}{2(N+2)} = \frac{1}{2}\left(\frac{3}{2} + \frac{1}{N+1} - \frac{5}{N+2}\right), \end{aligned}$$

as required. The coefficient of H_N vanishes because $A + B + C = 0$.

As $N \to \infty$, both $\frac{1}{N+1}$ and $\frac{5}{N+2}$ tend to 0, so

$$S_N \to \frac{3}{4}.$$

The sequence a_n

For $n \geq 2$, multiply the given ratios together:

$$\frac{a_n}{a_1} = \prod_{k=2}^{n} \frac{a_k}{a_{k-1}} = \prod_{k=2}^{n} \frac{k-1}{k+2} \times \prod_{k=2}^{n} \frac{2k-1}{2k-3}.$$

Ask yourself How do these two products simplify?

Write them out. The second product telescopes directly, because each numerator cancels the next denominator:

$$\prod_{k=2}^{n} \frac{2k-1}{2k-3} = \frac{3}{1} \cdot \frac{5}{3} \cdot \frac{7}{5} \cdots \frac{2n-1}{2n-3} = 2n - 1.$$

In the first product the numerators run through $1, 2, \ldots, n-1$ and the denominators through $4, 5, \ldots, n+2$. Everything from 4 to $n-1$ cancels, so

$$\prod_{k=2}^{n} \frac{k-1}{k+2} = \frac{(n-1)!}{(n+2)!/3!} = \frac{6}{n(n+1)(n+2)}.$$

The factorial form is valid for every $n \geq 2$, even when $n < 5$ and there is nothing to cancel. Therefore

$$\frac{a_n}{a_1} = \frac{6(2n-1)}{n(n+1)(n+2)}.$$

At $n = 1$ the right-hand side is $\frac{6}{6} = 1$, so the formula holds for all $n \geq 1$. As a check, $n = 2$ gives $\frac{18}{24} = \frac{3}{4}$, which agrees with the given ratio $\frac{1 \cdot 3}{4 \cdot 1}$.

When $a_1 = \frac{2}{9}$,

$$a_n = \frac{2}{9} \cdot \frac{6(2n-1)}{n(n+1)(n+2)} = \frac{4}{3} \cdot \frac{2n-1}{n(n+1)(n+2)}.$$

So a_n is $\frac{4}{3}$ times the nth term of the first series. Its partial sums are $\frac{4}{3}S_N \to \frac{4}{3} \cdot \frac{3}{4}$, and so

$$\sum_{n=1}^{\infty} a_n = 1.$$

Checks

- $N = 1$: the formula gives $\frac{1}{2}\left(\frac{3}{2} + \frac{1}{2} - \frac{5}{3}\right) = \frac{1}{6}$, which is the first term $\frac{1}{1\cdot 2\cdot 3}$. $N = 2$ gives $\frac{1}{2}\left(\frac{3}{2} + \frac{1}{3} - \frac{5}{4}\right) = \frac{7}{24} = \frac{1}{6} + \frac{1}{8}$. Both are correct.
- Numerically, the first 1999 terms sum to $0.74900\ldots$, approaching $\frac{3}{4}$ from below as the formula predicts ($S_N \approx \frac{3}{4} - \frac{2}{N}$).
- Generating a_n from the recurrence with $a_1 = \frac{2}{9}$, the first 299 terms sum to $0.9911\ldots$. The tail is about $\frac{4}{3} \cdot \frac{2}{300} \approx 0.009$, so this is consistent with a total of 1.

Common slips

- Mishandling the ends of the telescoping sum. Writing it in terms of H_N, as above, keeps the bookkeeping honest.
- Getting the factorial product wrong by one place. Write out the first and last few factors before cancelling.
- Forgetting to check that the formula for a_n/a_1 also holds at $n = 1$, where the product is empty.

Beyond STEP

Telescoping is the discrete version of the fundamental theorem of calculus: $\sum(f(n+1) - f(n)) = f(N+1) - f(1)$, just as $\int f' = f(b) - f(a)$. The second part is a *hypergeometric* sequence, in which the ratio of consecutive terms is a rational function of n. Gosper's algorithm decides exactly when such a sum has a closed form, and this is how computer algebra systems sum series. For the analysis of limits of partial sums, see Abbott, *Understanding Analysis*.

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